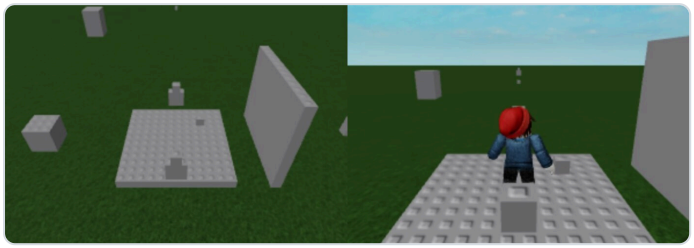
ROBLOX STUDIO EXPERT SKILLS · TEAM SKILL 4 OF 14

Networking and Security

Network programmer, anti-exploit engineer

`scenario-roblox-networking`

Treats every client call as attacker input: the server owns state, checks each remote in a fixed order, and proves it by firing hostile arguments from a playtest client.



Live capture, Studio 0.743: exploit test arena in Edit and from the playtest client. 24 live tests pass.



Server Authority Visualizer in a playtest: 100% instance prediction. Knight, Neeler, Hamilton, RDC 2026, 00:10:43.

USE IT WHEN

- Exploiters get free items, infinite money or duplicated pickups
- Remote spam, speed or teleport hacks, kill aura or hitbox abuse
- Designing remotes, choosing UnreliableRemoteEvent, or turning on Server Authority

WHAT THE EXPERTS INSIST ON

- NaN passes typeof(v) == 'number' and fails every comparison: check math.isfinite before any range check. Suphi Kaner (scripting educator)
- The client sends intent, never results: it names the item and the server prices it; a weapon reports origin, not damage. Roblox Creator Docs (client-server boundary)
- Roblox has no built-in lag compensation: validate hits against what the attacker saw, or use a cone of about 10 degrees. ev1 (two years on Server Authority)
- Test Server Authority with the visualizer under simulated latency, 60 ms with 10 ms jitter, and aim for 100 percent prediction. Knight and Neeler (Roblox, RDC 2026)
- When a player moves too fast, rubber-band them back instead of kicking: lag alone causes false positives. GnomeCode (tutorial creator)

WORKFLOW, WITH THE GATE BEFORE EACH NEXT STAGE

1. **Audit remotes and surfaces. Gate:** Zero errors; every warning fixed or justified; live tally equals offline.
2. **Validate and rate-limit. Gate:** No unvalidated handler, NaN-unsafe check, trusted value or missing rate limit.
3. **Check hits against history. Gate:** Out-of-reach, fake-origin and through-wall shots rejected; valid shots hit.
4. **Guard movement and ownership. Gate:** Honest walk 0 flags; client teleports rubber-banded; client CFrame writes move nothing.
5. **Fire exploits from the client. Gate:** Balances unchanged, no server errors, and the deliberately vulnerable control pays out.
6. **Detect, then act late. Gate:** One honeypot fire schedules enforcement; a flood alone stays below restrict.
7. **Test latency and bandwidth. Gate:** Rewind hit rate at least 0.8 under real ping; payloads under the cap.

KEY NUMBERS

UnreliableRemoteEvent payload cap	**1,000 bytes; live, 980 arrived and 1,000 was dropped**
Client-to-server requests per remote type	**about 500 per second per client**
NaN through a range-only check	**balance set to -9223372036854775808 (live)**
Moving target at 202 ms ping	**13 of 13 hits with rewind, 0 of 13 without**
Server Authority, client teleport attempt	**server moved 0 studs; honest 2 s walk moved 32.4**
Delay before visible enforcement	**random, 20 to 90 s**

CHECKS THE AGENT RUNS ON ITS OWN WORK

- Fuzz: every hostile case lands in args or rate counters; balances and health unchanged.
- 1,000 same-frame reward fires grant once; purchases never exceed the bucket.
- The console during the fuzz shows no error from your own handlers.
- Zero type errors on both Luau solvers, luau-lsp with a Rojo sourcemap.

HANDOFFS

Receives Remote list, folder layout and state ownership from architecture; hitbox and ability rules from gameplay.

Delivers Validated, finite, size-checked values to data; receipt-only grants to monetization; payload sizes to performance.

TOOLKIT `RBX_NET.PY`

- `audit_place()`: offline Lune audit of remotes, handlers and surfaces
- `fuzz()`: fires hostile arguments at a remote from the client
- `set_latency()`: simulated ping in each direction before Play

MAIN EXPERT SOURCES

Suphi Kaner
scripting educator, Stop Hackers / Exploiters: Roblox Scripting Tutorial

ev1
developer, two years on server-authoritative projects, Server Authority: Powering Competitive Gameplay, Inspire 2026

GnomeCode
tutorial creator, How to Secure your Game Against Exploiters

Jason Knight, Brian Neeler, Drew Hamilton
Roblox engineer, Roblox PM, studio developer, Mastering Server Authority, RDC 2026

STATUS

Tests 8 offline checks pass

In app 24 live tests pass, 1 skipped (needs two clients), in Roblox Studio 0.743

G2 networking and security

without skills	11/15
with skills	15/15